

FieldTrack Backend Migration

Node.js / Express !' Python / FastAPI

Module-wise Estimation — Single Developer, 23-Day Plan (Claude-Assisted)

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Basis & Critical Assumption

This plan fits the full 12-module migration into 23 working days (184 hours) with a single developer. This is only achievable if Claude (Claude Code) generates the first-draft code for each module — Pydantic schemas, route translation, SQL query porting, and pytest scaffolding — and the developer's time is spent primarily on review, testing, edge-case fixes, and integration rather than writing code from scratch.

- No team backup: a single person carries the full schedule with minimal slack.
- QA/testing buffer is compressed to 13 hours (vs. 56 hours in the unassisted estimate) — acceptable only because Claude also assists in porting the existing Jest test suite to pytest.
- Any blocking issue (illness, an unexpected bug, a Google Maps API discrepancy) will push the finish date, since there is no second developer to absorb it.

Module-wise Schedule (Sequential, Single Developer)

#	Module	Hours	Day Range
1	Core Infra & Setup (DB pool, JWT dependency, CORS/rate-limit, logging, migrate script)	12	Day 1-2
2	Auth (login, refresh, JWT issuance, bcrypt)	9	Day 2-3
3	Dealers (CRUD)	7	Day 3-4
4	Employees (CRUD)	7	Day 4-5
5	Assignments (dealer-rep assignment logic)	10	Day 5-6
6	Attendance (check-in/out, GPS tolerance, business-day boundary)	13	Day 6-8
7	Visits (lifecycle, haversine, idempotency, activity log)	25	Day 8-11
8	Navigation (route/turn-by-turn + Google Routes API)	15	Day 11-13
9	Geocode (Google Geocoding API)	10	Day 13-14
10	Notes / Reminders / Notifications	12	Day 14-16
11	Dashboard & Reports	11	Day 16-17
12	Follow-up Requests & Sync Failures	12	Day 17-18

Module subtotal: 143 hours (Day 1-18)

Final Phase — Integration, Deployment & Sign-off (Days 18-23)

Task	Hours	Day Range
Integration & parity testing (Claude-assisted comparison vs. ported Jest suite)	20	Day 18-21
Deployment migration (render.yaml, requirements.txt, Uvicorn config, cutover plan)	8	Day 21-22
Final review and production cutover	13	Day 22-23

Total

Metric	Value
Total effort	184 person-hours
Team size	1 developer (Claude Code-assisted)
Delivery window	23 working days
Scope change	None — logic, features, and UI remain unchanged
Risk level	High — no buffer, no backup developer, compressed QA